
API Security Risk Analysis Report - JSONPlaceholder 2026

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Executive Summary

I conducted a security analysis on the public test API <https://jsonplaceholder.typicode.com> using Postman. The purpose was to identify common API vulnerabilities that exist in real-world applications. Several high and medium risks were found, particularly around authentication and data exposure.

Methodology

- Used Postman to send various HTTP requests (GET, POST).
- Tested without any authentication or rate limiting.
- Analysed responses for data exposure and security controls.

Key Findings

- No Authentication Required (High Risk) Anyone can read and create data without login or API keys. In a real system this could lead to full data breach.
- No Rate Limiting Visible (Medium Risk) The API does not appear to limit the number of requests, making it vulnerable to abuse or denial-of-service attacks.
- Full Data Exposure (High Risk) Responses return complete user and post data without any filtering or masking of sensitive fields.
- No Input Validation on POST (Medium Risk) The API accepts any data without strict validation.

Risk Classification Table

FINDING	RISK LEVEL	BUSINESS IMPACT
No Authentication	High	Complete data exposure
No Rate Limiting	Medium	Possible abuse and service disruption
Full Data Exposure	High	Privacy and compliance violations
Weak Input Validation	Medium	Risk of injection attacks

Remediation Recommendations

- Implement proper authentication (OAuth 2.0 or API keys).
- Add rate limiting (e.g., 100 requests per minute per user).
- Use proper data filtering and never expose sensitive information.
- Validate and sanitise all inputs on the server side.
- Enable logging and monitoring for suspicious activity.

Conclusion

This task gave me practical experience in testing API security. Real-world APIs must have strong authentication, rate limiting, and input validation to protect company and customer data.